Swarnima Jaiswal

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Summary

AI/ML-focused B.Tech student with hands-on experience in Machine Learning, Deep Learning, Generative AI, NLP, and Large Language Models. Built AI-driven projects including a Financial RAG Chatbot and an AI-powered Interview Assistant using Python, FastAPI, RAG, LLMs, and modern AI frameworks. Strong foundation in AI/ML concepts and practical implementation, with a focus on developing intelligent solutions for real-world problems.

Education

Vellore Institute Of Technology 2023 - 2027
B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning)
CGPA: **8.60/10** | Full-time

Class XII 2021 - 2022
Children Senior Secondary School Board: CBSE
Percentage: **72.8%** | Full-time

Class X 2019 - 2020
Children Senior Secondary School Board: CBSE
Percentage: **87.4** | Full-time

Technical Skills

Programming Languages: Python, SQL, Java
AI & Machine Learning: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)
Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, LangChain, NumPy, Pandas, Streamlit
Backend & APIs: FastAPI
Developer Tools: GitHub,VS Code,PyCharm,Jupyter Notebook, Google Colab

Projects

Financial RAG Chatbot | *Python, QLoRA, RAG, Hugging Face, Vector Search* July 2026

- Developed a finance-focused **Retrieval-Augmented Generation (RAG) chatbot** for document-based question answering using **Qwen2.5-1.5B-Instruct**
- Implemented the complete RAG workflow covering**5 key stages:** document preparation, text chunking, embeddings, vector retrieval, and LLM-based response generation.
- Integrated **QLoRA** fine-tuning and RAG workflows to compare base, fine-tuned, and retrieval-augmented LLM configurations.
- Evaluated model performance on **FinanceBench,** canalyzing the impact of retrieval and strict numeric prompting on financial question-answering performance.
- Worked with **3 evaluation configurations** — base model, fine-tuned model, and RAG-based approach — to study the effect of retrieval on financial QA.

AI Interview Assistant | *Python, FastAPI, React.js, Vite, Groq, GPT-OSS-20B, PyMuPDF* May 2026

- Developed an AI-powered interview platform that combines**resume parsing, LLM-based question generation, automated evaluation, and performance analytics** in a single workflow.
- Built **5+ FastAPI REST API capabilities** covering PDF resume parsing, interview sessions, question generation, answer submission, and AI-based evaluation.
- Implemented adaptive interviewing using **4 contextual inputs** — resume information, previous answers, evaluation results, and previously asked questions — to generate subsequent technical questions.
- Designed an automated evaluation system across **4 scoring dimensions:** technical accuracy, clarity, depth, and relevance.
- Built a performance dashboard covering **5+ analytics outputs,** including total score, average score, category-wise performance, strengths and weaknesses, and question-wise results.

Extra-Curricular Activities

- **Beyond Level-O-Hackathon | Team Lead & Presenter |** Certificate Aug 2024
  Collaborated in a **4-member multi-skilled team**, contributed to technical discussions, and presented the final prototype to judges, demonstrating strong **teamwork, leadership, and communication** skills.
- **Think & Code | Individual Participant |** Certificate Aug 2026
  Participated individually in **Think & Code**, representing **VIT Bhopal University** at the technical event organized by **MKSSS's Cummins College of Engineering for Women (CCOEW), Pune**.
- **Adobe University Hackathon | Participant |** Certificate Aug 2026
  Participated in the **Adobe University Hackathon**, organized by **Adobe**, gaining exposure to collaborative technical problem-solving.

Certifications

- **Applied Machine Learning in Python | University of Michigan** Dec 2025 Certificate
- **AWS Certified Cloud Practitioner — Official Practice Question Set (CLF-C02)** Sep 2025 Certificate